REVIEW CANDIDATE · 26 OCTOBER 2026

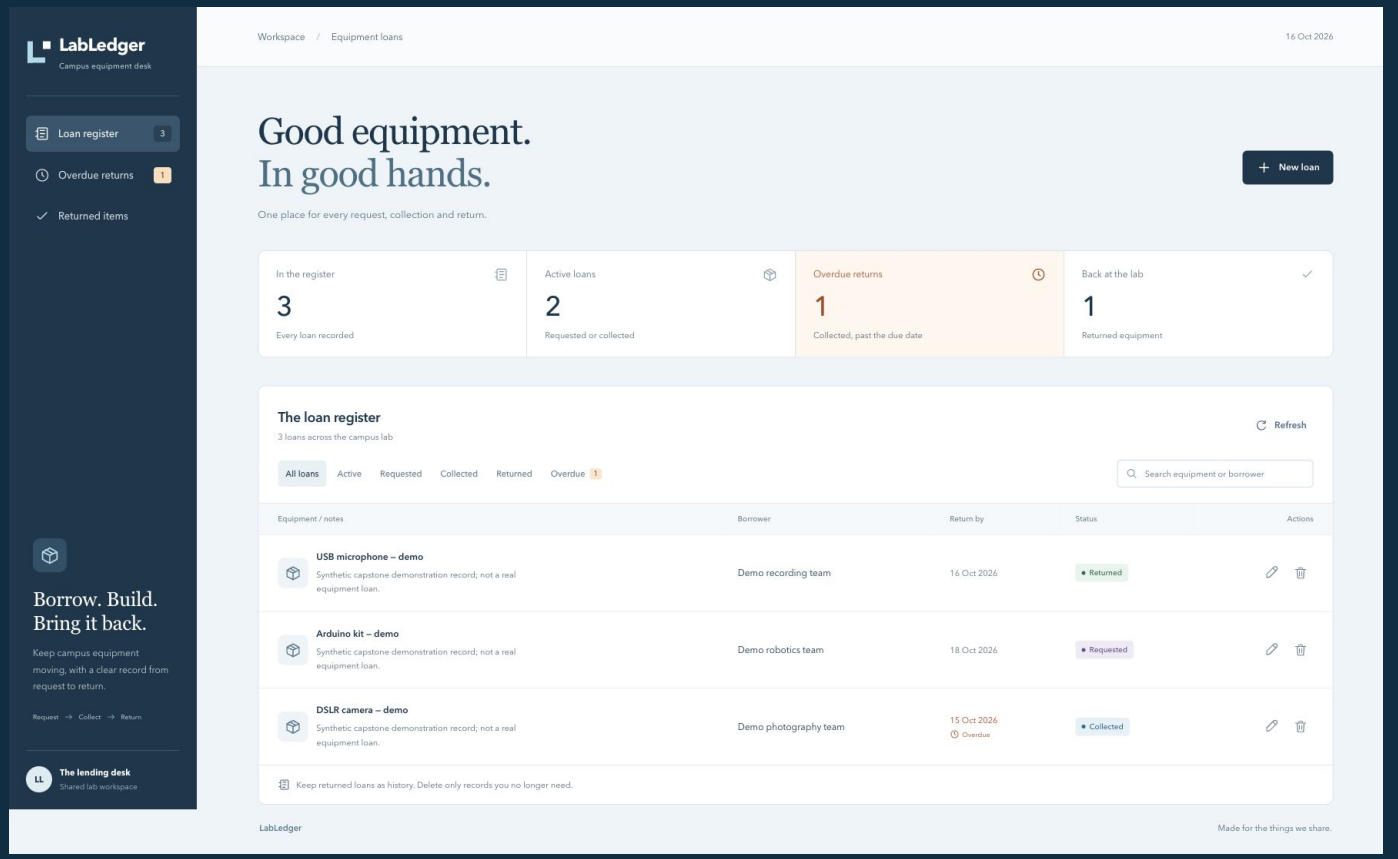
LabLedger

Shared equipment. Clear returns.

React · FastAPI · PostgreSQL
Docker · CI · Helm · Monitoring

VERIFIED ON EKS

Temporary cloud demo cleaned up.



One clear equipment-loan lifecycle

Request → collect → return

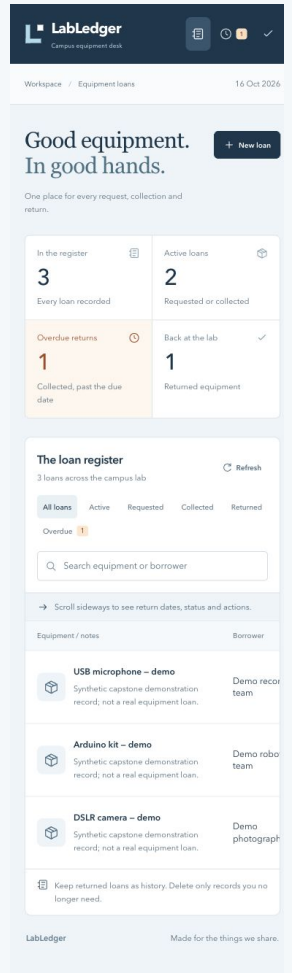
A lab desk records the borrower, equipment and return date.

Overdue has a precise meaning

Collected + due date before today in Asia/Kolkata. Requested loans are active, but not overdue.

Usable on small screens

Search, status filters, readable actions and an accessible dialog. Demonstration records are explicitly fictional.



Genuine School-profile mobile capture

Follow a request through the stack

React UI
Relative /api calls

→

Nginx / Ingress
/ and /api routes

→

FastAPI
Validation + metrics

→

PostgreSQL
Alembic + loans

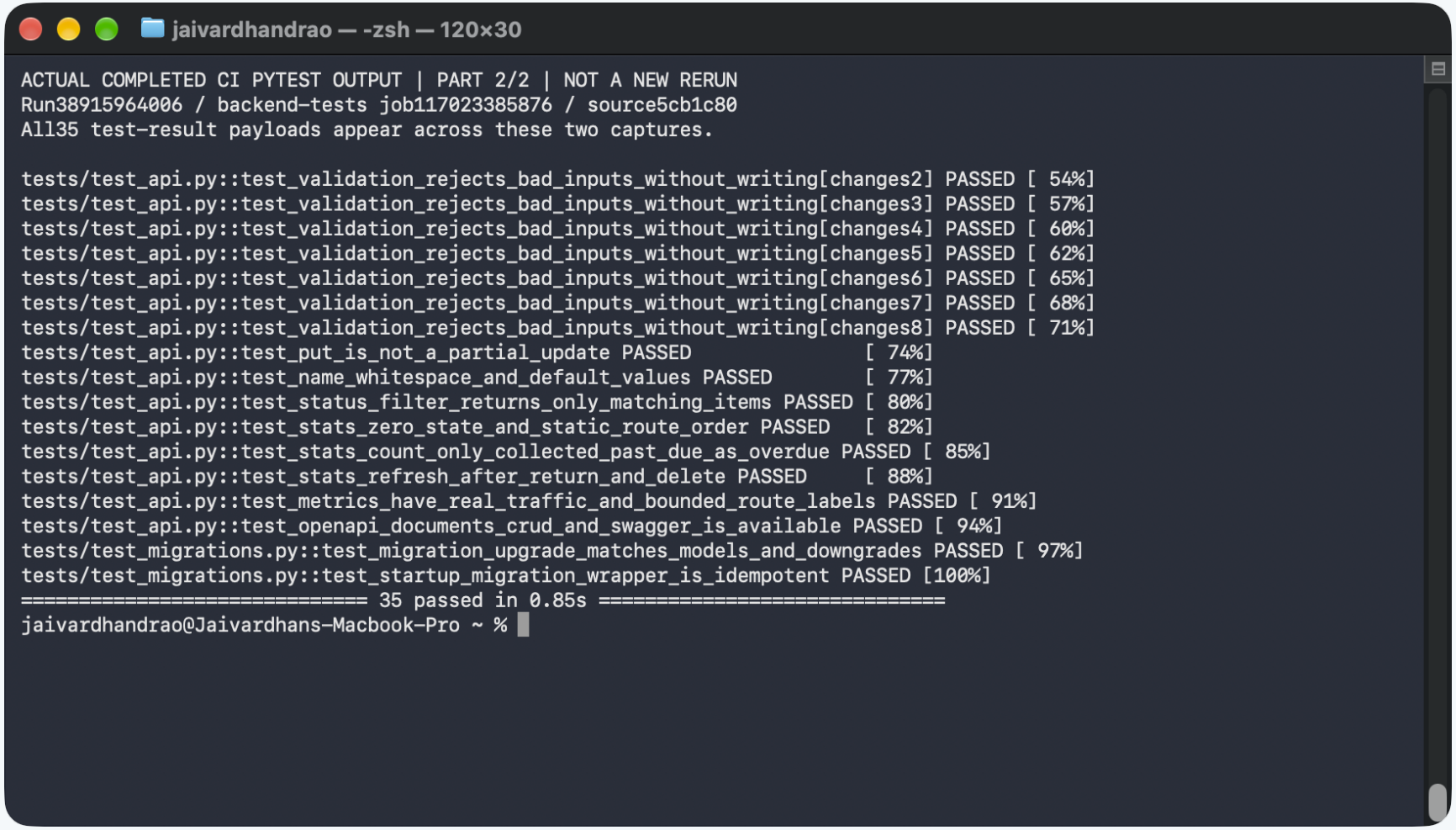
Compose localhost:3000 → frontend → backend → database

Kubernetes localhost:8088 → Ingress → ClusterIP Services → Pods

Prometheus → Grafana

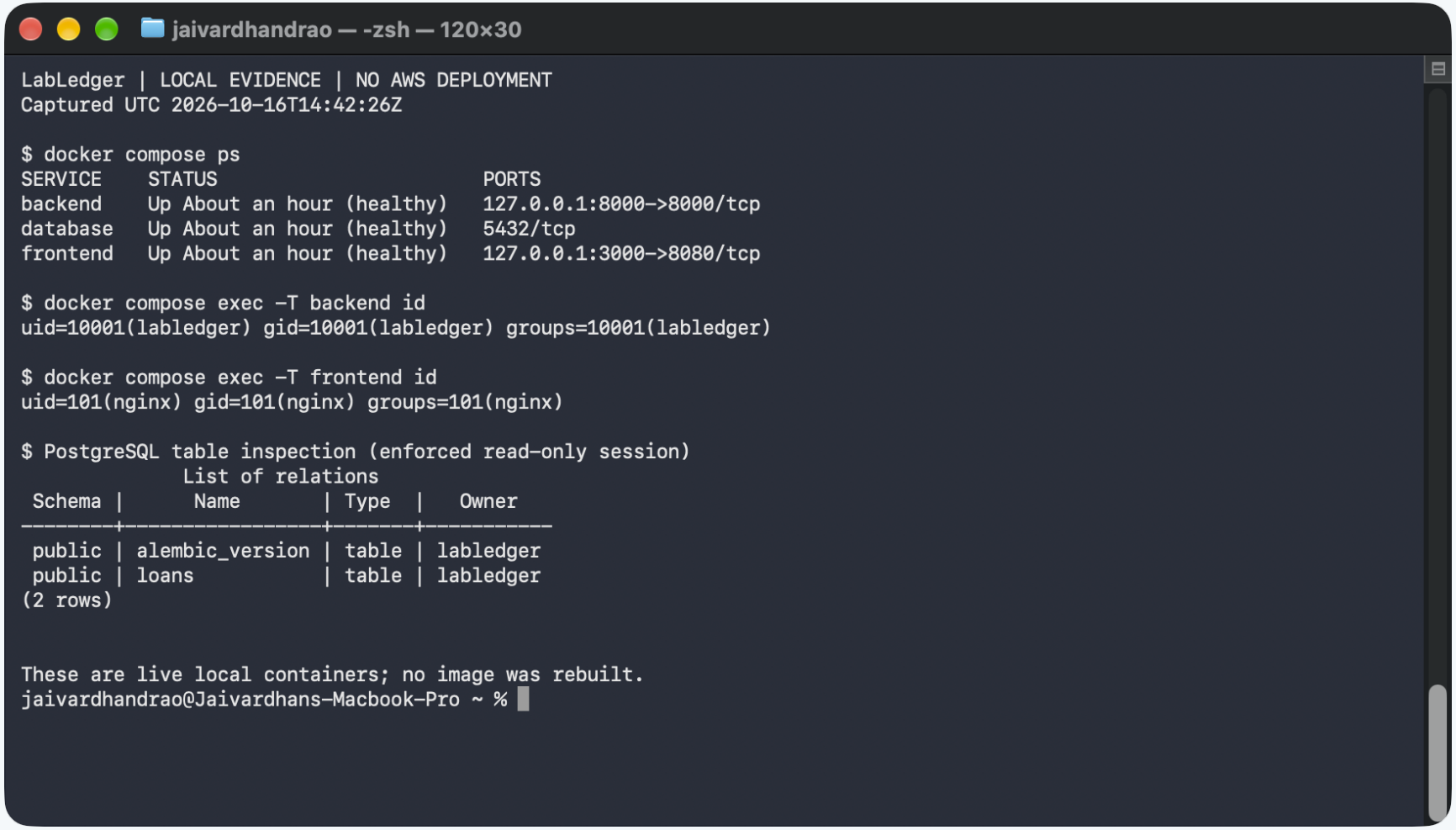
/metrics on both API replicas; real HTTP counters and latency histograms.

35 isolated tests, plus real-stack checks



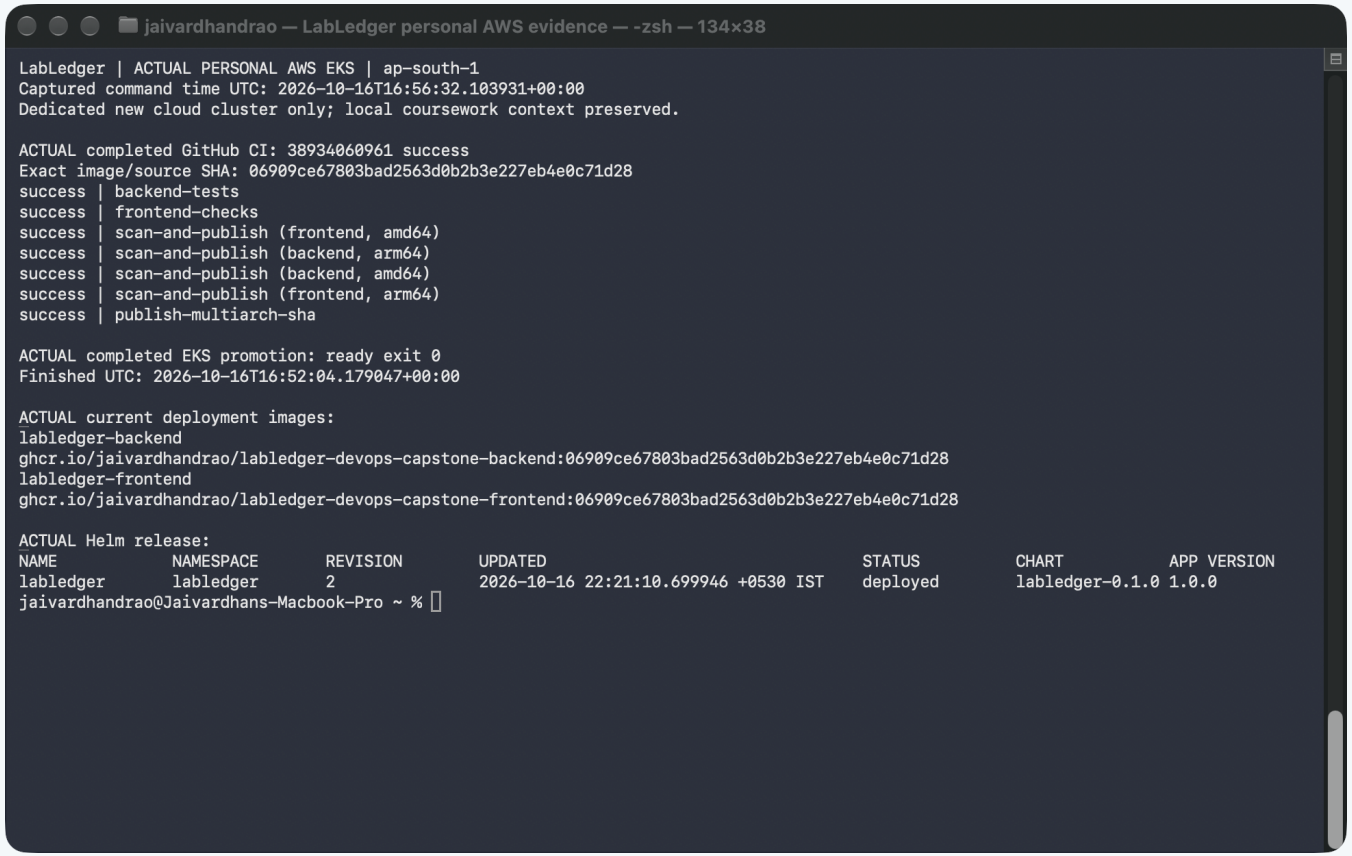
Actual completed CI pytest output; all 35 results are available across the two native captures.

Three healthy services; two non-root images



Live local containers and enforced read-only table inspection; original volumes preserved.

A real change reached the running application



Shown: historical 06909ce EKS rollout, release 2; infrastructure is retired.

Later local: f271a837 → 7 green CI jobs → exact images → Helm release 4.

Open the final local change’s seven-job green run

The scan gate found a problem, then verified the repair

Before

Strict scan failed

Debian runtime findings were preserved in the original report.

After

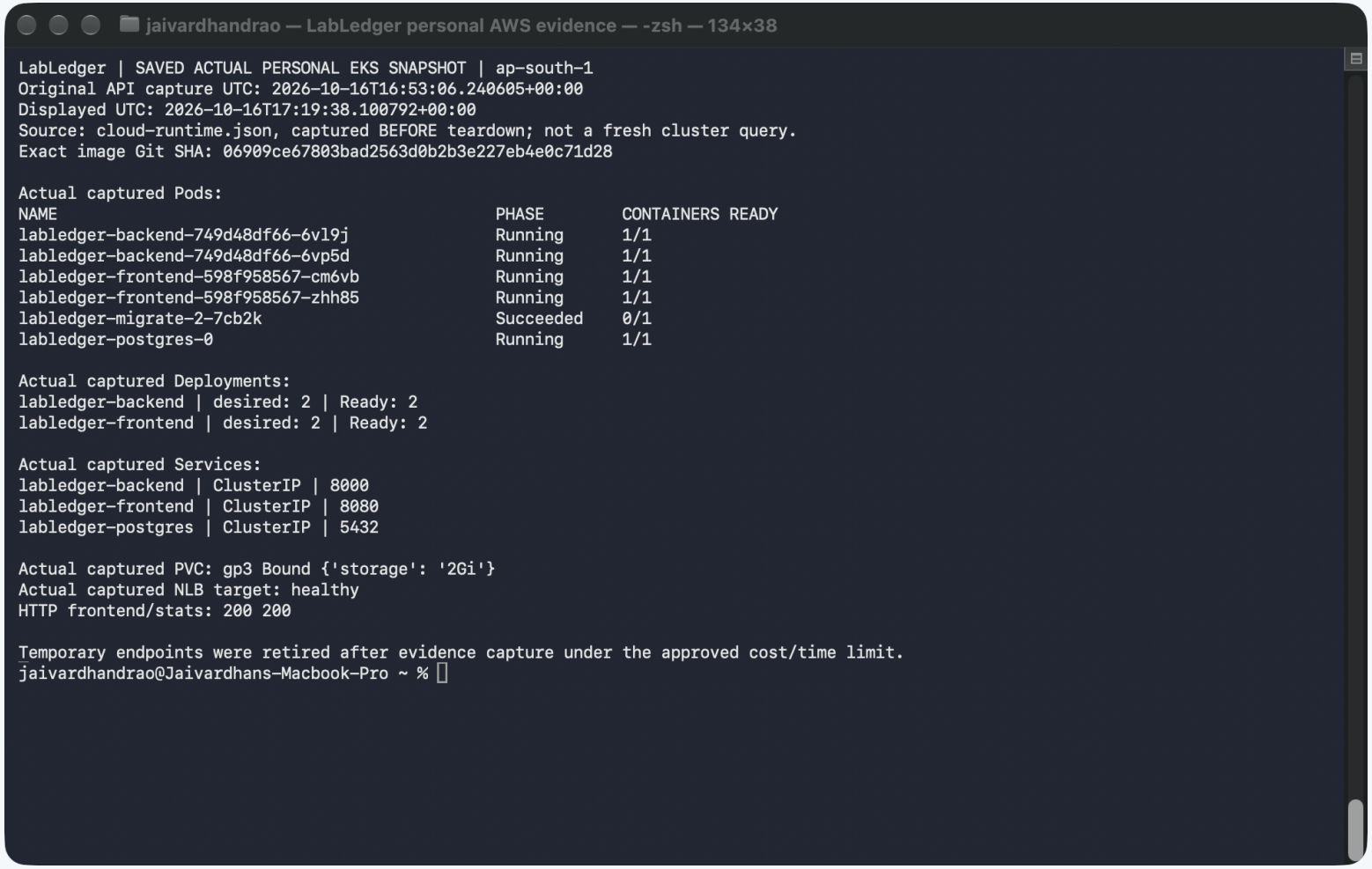
4 strict gates passed

Backend + frontend
amd64 + arm64
No HIGH / CRITICAL allowed

Gate: --severity HIGH,CRITICAL --ignore-unfixed=false --exit-code 1

A clean scan is evidence about that image at that time; it is not a security guarantee.

Helm release 2 ran on the dedicated EKS cluster



Actual EKS: ClusterIP Services, bound gp3, completed migration and deployed Helm releases.

Resilience was exercised, not only configured

2 → 4 Ready → 2

Observed cloud API HPA response; one worker throughout.

7,431 reads · 12 failures

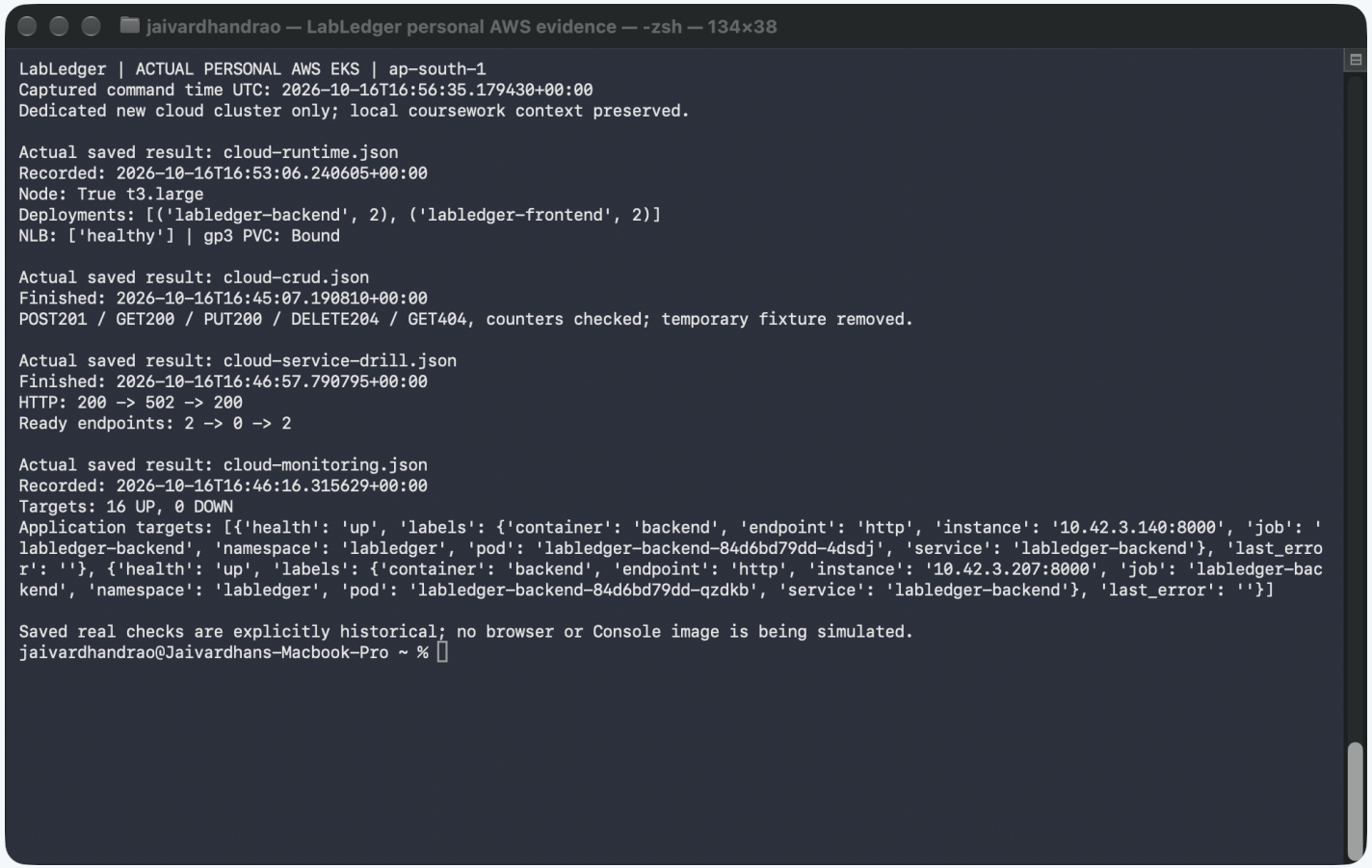
Separate lower-rate recovery: 40/40 HTTP 200. Burst failures retained.

200 → 502 → 200

Service selector fault, diagnosis and real recovery.

Actual cloud records retained. Earlier local 2,338 reads / 0 failures are separate evidence.

Monitoring evidence has a clear scope

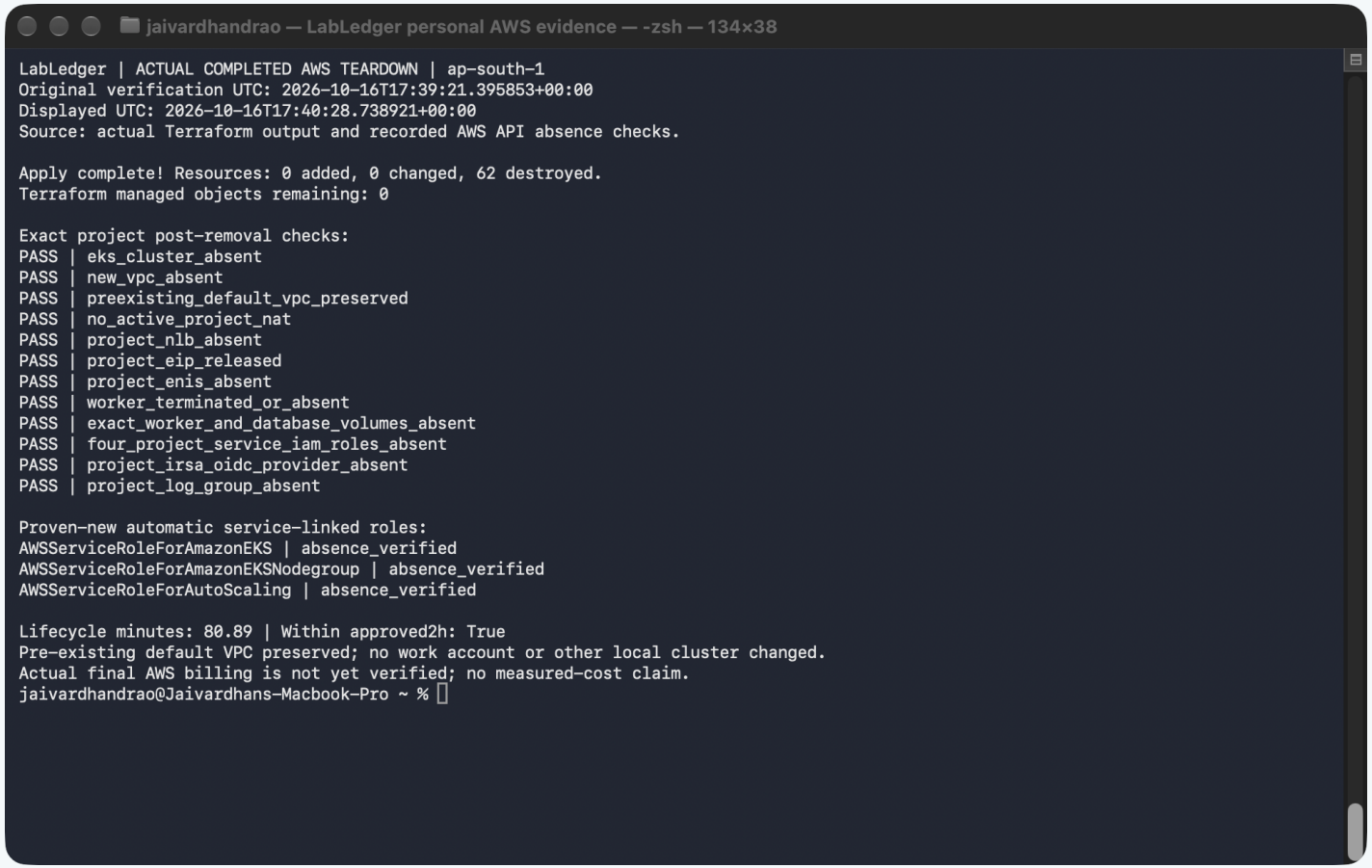


Cloud: 16/16 targets UP at capture time; actual CRUD and Service recovery retained.

Local Grafana: six populated panels captured privately on 16 Oct, 23:03 UTC.

One-hour history spans revisions and recovery; private browser image is not embedded.

Real AWS deployment, then verified cleanup



62 created → real runtime verified → 62 destroyed; pre-existing resources preserved.

Planning estimate about US$2.43 before tax; actual final AWS bill is not yet verified.

Present the verified work and its limits

Prepared locally for review

Updated deck + actual local change-to-rollout video under demo/. Nothing new is published.

Full presentation remains incomplete

ArgoCD is absent. The slides-first AWS Console / GitOps / browser walkthrough is not recorded.

Keep the evidence honest

Local populated Grafana capture is private historical evidence. AWS is retired; the form is user-reported submitted.

Submitted repository; prepared review files are not yet published